Project Report

Global Fuel Prices and Taxation Trends (2020–2026)

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1. Introduction

1.1 Project Overview

This individual project explores global fuel prices and taxation trends from January 2020 to April 2026 using Tableau. The analysis focuses on petrol, diesel, LPG, Brent crude prices, tax percentage, regions, countries, and income levels where those fields are available.

1.2 Purpose

The purpose is to make historical patterns and cross-country differences easier to explore through visualizations, an interactive dashboard, and a data story.

2. Ideation Phase

2.1 Problem Statement

Stakeholders need a clear way to compare fuel prices and related indicators across countries, regions, income levels, and time. Charts can support exploration, but should not be used to claim causation without further evidence.

2.2 Empathy Map Canvas

The three stakeholder perspectives are an energy policy consultant, a global supply chain manager, and an energy economics researcher. Their needs are respectively policy comparison, logistics/procurement context, and careful exploration of price-tax-income relationships.

2.3 Brainstorming

• Trend analysis for petrol and LPG.

• Country comparisons for petrol, LPG, and tax percentage.

• Regional LPG treemap and petrol-versus-Brent trend.

• Income-level comparison and tax-versus-petrol scatter plot.

• Interactive dashboard and story.

3. Requirement Analysis

3.1 Customer Journey Map

Explore question → select relevant dashboard filters → compare charts and time trends → interpret patterns with limitations → use findings to decide what needs further investigation.

3.2 Solution Requirement

• A dataset with date, country, region, fuel prices, tax percentage, crude oil, and income-level information where available.

• Readable visualizations with clear labels, units, and aggregation.

• Interactive country/region filters, subject to final testing.

• A dashboard, story, public link, and supporting documentation.

3.3 Data Flow

Dataset → Tableau field review and preparation → ten worksheets → dashboard and story → Tableau Public → GitHub and final submission.

3.4 Technology Stack

Tableau Desktop 2026.3.0, Tableau Public (published status reported by the student; link to verify), GitHub, and Word/PDF documentation.

4. Project Design

4.1 Problem-Solution Fit

The existing charts provide views of historical trends, country differences, regional distribution, tax-price association, crude oil comparison, and income-level variation that align with the three stakeholder scenarios.

4.2 Proposed Solution

The solution is the existing Tableau workbook with ten worksheets, “Global Fuel Prices Dashboard,” and “Global Fuel Prices – Story.”

4.3 Solution Architecture

The data is loaded into Tableau, reviewed and used to create worksheets, which are combined into a dashboard and story for exploration and publication.

5. Project Planning and Scheduling

The work is organized into dataset review/preparation, worksheet creation, dashboard/story assembly, publishing and repository checks, testing, documentation, and demonstration preparation. Actual sprint dates and story-point completion are not confirmed.

6. Functional and Performance Testing

Final checks should verify that worksheets open without errors, dates and units are sensible, country/region filters update the intended charts, dashboard layout remains readable, story points work in sequence, and the published page is accessible. Record actual test outcomes before submission.

7. Results

The confirmed deliverables are ten worksheets, a dashboard, and a story. Numerical findings are intentionally not stated here because the actual chart values and conclusions have not been provided for verification.

7.1 Output Screenshots

Insert genuine screenshots from Tableau: dashboard, story, and each required worksheet. Do not use illustrative or fabricated screenshots.

8. Advantages and Disadvantages

• Advantages: visual comparison of trends; country/region exploration; dashboard and story support communication.

• Limitations: conclusions depend on dataset quality and field definitions; visual association does not establish causation; subsidy effects and forecasting are not confirmed as measured/implemented.

9. Conclusion

The project organizes the supplied fuel-price dataset into Tableau views designed to support exploration of historical trends and cross-country differences. Final conclusions should be written after verifying the charts and their values.

10. Future Scope

• Validate data quality and document all field definitions.

• Add a geographic heat map or KPI cards only if required by the official rubric and supported by the data.

• Investigate subsidy effects only if suitable subsidy data and methods are available.

• Consider a forecasting model only as future work unless it is built and validated.

• Improve the repository README and add verified public/demo links.

11. Appendix

Dataset

• Reported dataset size: 27,468 records, 84 countries, 7 global regions, 10 columns, approximately 2.7 MB.

• Reported date coverage: January 2020 to April 2026.

• Fields confirmed from the handover: Date, Country, Region, Income Level, Lpg Usd Liter, Brent Crude Usd, Tax Percentage, Diesel Usd Liter, and a petrol field displayed in Tableau as AVG(Petrol used liter). Confirm the exact petrol field spelling in the data pane before final submission.

• The tenth column has not been identified in the handover; do not assume its name.

Existing Tableau Visualizations

Worksheet	Business question
LPG Price Trend	How has LPG price changed over time?
Petrol Price Trend	How has petrol price changed over time?
LPG Price by Country	How do LPG prices compare across countries?
Tax Percentage by Country	How does tax percentage vary by country?
LPG Price by Region (treemap)	How are LPG prices represented across regions in the treemap?
Fuel Price vs Crude Oil Trend	How do the displayed petrol and Brent crude price trends move over time?
Petrol Price by Country	How do petrol prices compare across countries?
Fuel Price Comparison Trend	How do the fuel-price trends shown in the worksheet compare over time?
Income Level vs Fuel Price	How do fuel prices vary across income-level groups in the available data?
Tax vs Petrol Price	What association, if any, is visible between tax percentage and petrol price?

Links

Tableau Public URL: to be added after verification. GitHub repository URL: to be added after verification. Project demonstration link: to be added if required.